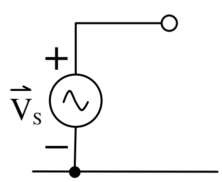
Equivalent circuits are an indispensable tool for modeling/analyzing power systems and for making power system calculations easier to perform. An equivalent circuit is a special type of mathematical model of an actual 1ϕ or three 3ϕ power system which linearly approximates the combined electrical behaviors of the power system components. It is important to always remember that there is a difference between the actual power system and its equivalent circuit. The actual power system is something real that can physically exists. However the equivalent circuit is only an abstract mathematical model that is created and used to approximate the behavior of the real world power system and to make power system calculations easier to perform.

Equivalent circuits for power systems are created by assuming that each type of power system component (**Sources**, **Loads**, **Interconnects**) have known equivalent circuit models. These individual component models can then be used as simple building blocks to create a composite equivalent circuit for the power system.

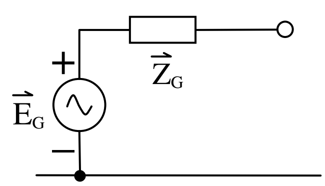
Sources in power systems can be AC generators, ties to an electric utility company ties, or ideal sources. All power system sources have one thing in common. They are what produce the nominal operating voltage levels in a power system. Therefore their equivalent circuit models are all based on the concept of an ideal voltage source. An ideal voltage source in power systems is also called an **infinite bus** because of its ability to theoretically supply an infinite amount of power.

Accurately modeling and predicting the behavior of AC generators is very complicated and requires a detailed knowledge of the physical construction and internal operation of this special type of rotating electric machine. This is best done in a course on electric machinery or an advanced power system operation course, and not in this course. Accurately modeling and understanding the behavior of electric utility company ties is also very complicated, and will not be covered in this course. Fortunately there are simplified equivalent circuit models that we can use in this course to approximate the behavior of generators and electric utility company ties. Shown below are the different equivalent circuit models we will use in this course for power system sources.

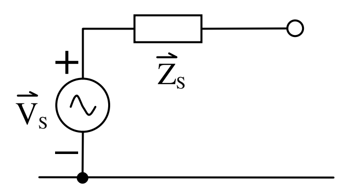
Ideal AC Source Model



AC Generator Model

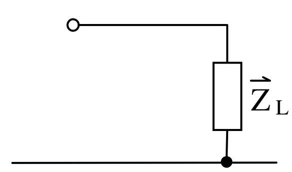


Electric Utility Tie Model



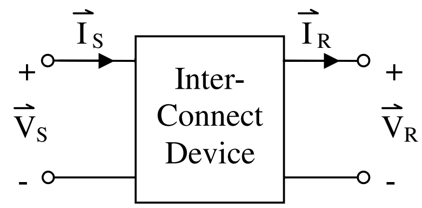
Loads in real world power systems are devices like motors, light bulbs, computers, household appliances, heavy industrial equipment, etc. No matter what the loads actually physically are, they can be modeled in an approximate way using simple equivalent impedances. The equivalent circuit model for a load is shown below.

Load Model



Interconnects in real world power systems are the components that connect the sources to the loads. Interconnect components include switches, circuit breakers, fuses, insulated conductor power cables, open conductor overhead transmission lines, and transformers. A special equivalent circuit model, called the general 2-port network model, for any interconnect component is shown below.

General Interconnect Component Model

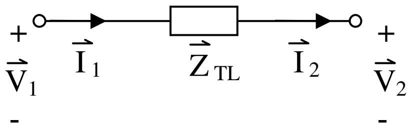


This general 2-port network model has both an input port and an output port. Recall from basic circuit theory that a port is a pair of terminals where both the voltage between the terminals and the current passing through the terminals can be defined. Interconnect component models have 2-ports, butpower system source and load models have only 1-port and are consequently called 1-port networks.

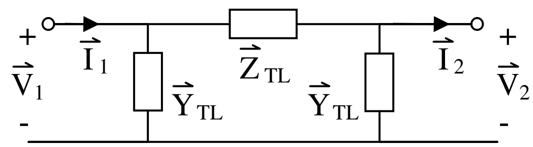
In this course, much time will be spent discussing and using special 2-port network models for open conductor overhead transmission lines, insulated conductor power cables, and transformers. Models for these specific interconnect components vary greatly in their type and their mathematical complexity. One of the main purposes of this course is to learn how these models are derived from the physical design and construction details of the device, and to learn how to use these models to perform system and device calculations.

For **open conductor overhead transmission lines** and **insulated conductor power cables**, two different equivalent circuit models are commonly used. These two models are called the ***Series Impedance Model*** and the ***Pi Model*** and are shown below.

Series Impedance Model

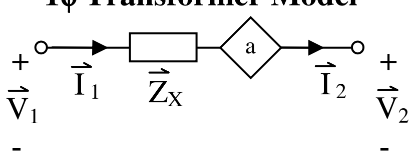


Pi Model

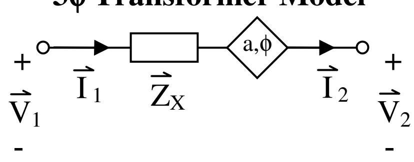


For **transformers**, there are four different equivalent circuit models that are commonly used. However in this course we will use only one of these models, which is most often called the *Simplified Steinmetz Model* or just the *Simplified Model*. In this course we will just call it the *Transformer Model*, since we are not going to use any of the other models. There are two different variations of this transformer model that we will specifically use. One is for modeling 1φ transformers, and the other is for modeling 3φ transformers. Both are shown below.

1ϕ Transformer Model



3ϕ Transformer Model



Now that the equivalent circuit models for power system components are defined, then the equivalent circuit for any power system can be built. Power system equivalent circuits should always be drawn with an extra long reference line at the very bottom. This line is the reference node in the circuit, and all voltages in the circuit are defined with respect to this reference node. An important rule to remember when creating an equivalent circuit for a power system is to never draw anything below the reference node line. It is recommended that this line be drawn first before the rest of the circuit is drawn to emphasize its importance. Another important rule that should always be followed is to always draw sources and load impedances vertically with their bottom terminals connecting to the reference line in the circuit.

Creating an equivalent circuit for a power system can be done very methodically using the step-by-step procedure documented below.

1. Draw an exaggeratedly long horizontal reference line, leaving adequate room above this line for drawing the rest of the circuit. This reference line will be a common node that all voltages are defined with respect to.

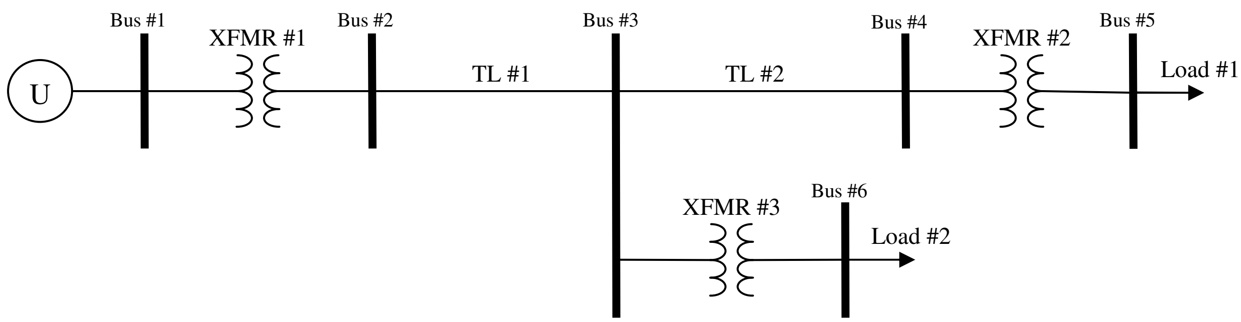
2. Vertically draw in the equivalent circuits for all of the sources (ideal sources, generators, and electric utility ties) in the system, connecting their bottom terminals to the reference line, labeling their parameters, and labeling their top terminals with numbers corresponding to busses in the actual power system.

3. Vertically draw in the equivalent circuits for all of the loads in the system, connecting their bottom terminals to the reference line, labeling their parameters, and labeling their top terminals with numbers corresponding to busses in the actual power system.

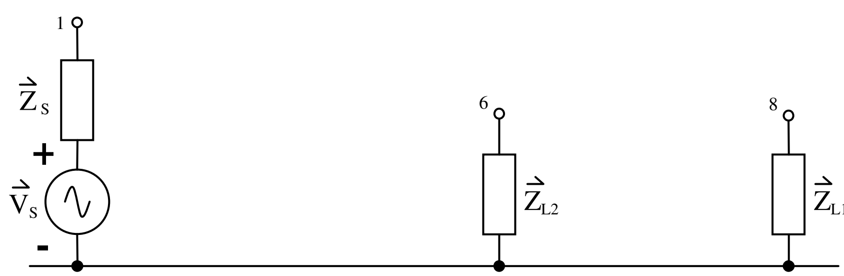
4. Horizontally draw in the equivalent circuits for all of the interconnect components in the system, connecting them to already existing nodes, and labeling their parameters. Additional nodes should be added to the circuit and labeled for busses in the system that have only interconnect component connections and no source or load connections.

5. Redraw the circuit if necessary to eliminate or at least minimize crossover lines.

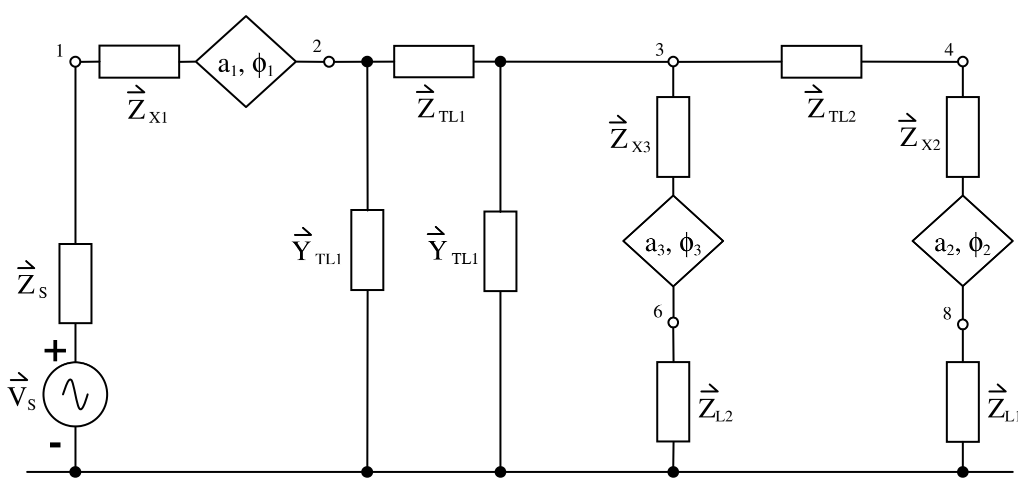
Example #1:
A one-line diagram for a 3ϕ power system is given below. Using the step-by-step procedure just documented, draw the L-N equivalent circuit for this 3ϕ power system. Use the pi-model for transmission line TL#1 and the series impedance model for transmission line TL#2.



First, steps 1, 2, and 3 of the procedure are combined and a partial equivalent circuit is shown below.



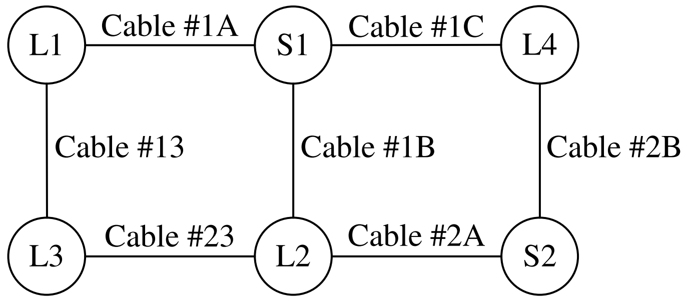
Next, step 4 of the procedure is done to yield the final equivalent circuit shown below.



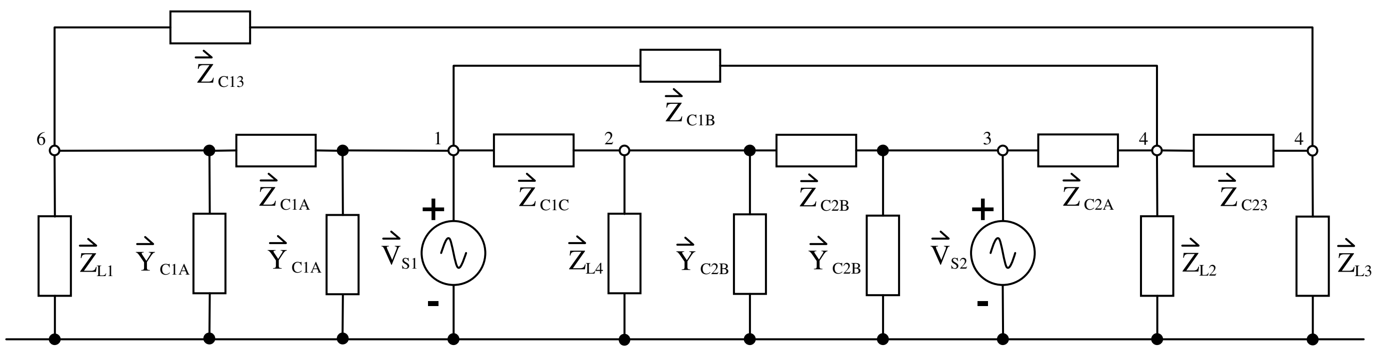
Step 5 is not required in this case since the equivalent circuit from Step 4 is already very clear and neat.

Example #2:

A block diagram for a 1ϕ power system is given below. In this system, two distributed sources supply power to four distributed loads through a network of power cables. Using the step-by-step procedure just documented, draw the equivalent circuit for this 1ϕ power system. Use the pi-model for power cables #1A and #2B, and use the series impedance model for the remaining power cables.



The resulting equivalent circuit shown below for the system is obtained by applying the step-by-step procedure.



Notice in the previous example that an intermediate equivalent circuit for steps 1, 2, and 3 of the procedure was not first drawn, as was done in the first example. This is the recommended practice and what you should do in all future problems. It is too time consuming and not necessary to redraw equivalent circuits multiple times just to illustrate the steps of the procedure that was used to draw them. So do not do this.

The only time an equivalent circuit should be redrawn is if it is illegible, unclear, or contains crossover lines and can be redrawn to eliminate these crossover lines. Remember that in the 5-step PSA procedure, the purpose of drawing an equivalent circuit for a power system is so that circuit analysis and calculations can be done. Therefore if an equivalent circuit is drawn in a systematic, neat, and clear way then it can be analyzed easier and other calculations can be done easier.